

Multiple Choice (10 marks)

1. Which of the following compounds has a ^1H resonance approximately 1.55 kHz away from TMS on a spectrometer with a 12.0 T magnet?
 - (a) CH_3F
 - (b) CH_3Cl
 - (c) CH_3Br
 - (d) CH_3I
2. Calculate the Q-factor for an X-band EPR cavity with a resonator bandwidth of 1.58 MHz.
 - (a) $Q = 1012$
 - (b) $Q = 2012$
 - (c) $Q = 3012$
 - (d) $Q = 6012$
3. Which of the following is required for both paramagnetism and ferromagnetism?
 - (a) Strong oxidizing conditions
 - (b) Low-spin electron configuration
 - (c) Metallic physical properties
 - (d) Unpaired electrons
4. Which nuclide has an NMR frequency of 115.5 MHz in a 20.0 T magnetic field?
 - (a) ^{17}O
 - (b) ^{19}F
 - (c) ^{29}Si
 - (d) ^{31}P
5. Cobalt-60 is used in the radiation therapy of cancer and can be produced by bombardment of cobalt-59 with which of the following?
 - (a) Neutrons
 - (b) Alpha particles
 - (c) Beta particles
 - (d) X-rays

True / False (10 marks)

Answer True or False

6. True or False: T_2 , unlike T_1 , is sensitive to molecular motions at the Larmor frequency, which impacts relaxation times in NMR spectroscopy.
7. True or False: A spontaneous process is always exothermic and releases heat to the surroundings.
8. True or False: Planck's quantum theory successfully explains the blackbody radiation curves observed in experiments.
9. True or False: The hydrogen atom in $CHCl_3$ does not undergo rapid intermolecular exchange, leading to multiple peaks in its 1H spectrum.
10. True or False: The limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule is $3.5 R$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the thermal stability of hydrides of group-14 elements, analyzing the factors that influence their stability and arranging them in order from lowest to highest stability.
12. Discuss the significance of the absorption of electromagnetic radiation at 190 nm by proteins, specifically focusing on the implications of $\pi \rightarrow \pi^*$ transitions in the ultraviolet region of the spectrum.

End of Paper